GARCH and stochastic volatility models

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Chapter 4: Multivariate GARCH models

1. Motivations

2. Various Multivariate GARCH
   - Direct modelling of $\boldsymbol{H}_t$
   - Conditional correlation models
   - Factor models

3. Existence of stationary solutions

Why introducing multivariate GARCH models ?

- Similar to the VARMA extension of ARMA models

  However, the extension is more problematic, as there is not a unique formulation of MGARCH models.
- Modeling the comovements of financial series is of practical importance

Why introducing multivariate GARCH models ?

The conditional variance-covariance matrix appears naturally in several areas of finance, for instance :

- Portfolio choice
- Time-varying betas
- Conditional risk of a portfolio

Multivariate GARCH

- In the last 20 years : plethora of extensions of the univariate GARCH

$$\epsilon_t = H_t^{1/2}\eta_t, \quad \text{with} \quad (\eta_t) \text{ iid } (\mathbf{0}, I)$$

Reviews on the literature on MGARCH : Bauwens, Laurent and Rombouts (2006), Silvennoinen and Teräsvirta (2009), Bauwens, Hafner and Laurent (2012), Francq and Zakoïan (2019, Chapter 10).

- None seems to prevail, all have their particular strengths and weaknesses.
- Problems : existence of solutions, numerical complexity...

What can we expect from a specification of H_t ?

- Easy conditions for positive-definiteness of $\boldsymbol{H}_t$
- Explicit stationarity conditions
- Stability by aggregation ($P\epsilon_t$ in the same class for any P)
- Sufficiently flexible to capture the dynamic properties of financial series
- Parsimony in terms of parameters
- Avoid excessive inversion of the conditional variance (required in the numerical optimization of the likelihood for all t and every iteration)

Diagonal model

Bollerslev, Engle, Woolridge (1988)

$$
\begin{aligned}
H_t &= \begin{bmatrix} \omega_{11} & \dots & \omega_{1m} \\ \vdots & & \vdots \\ \omega_{1m} & \dots & \omega_{mm} \end{bmatrix} + \sum_{i=1}^{q} \begin{bmatrix} a_{11}^{(i)} \epsilon_{1,t-i}^2 & \dots & a_{1m}^{(i)} \epsilon_{1,t-i}\epsilon_{m,t-i} \\ \vdots & & \vdots \\ a_{1m}^{(i)} \epsilon_{1,t-i}\epsilon_{m,t-i} & \dots & a_{mm}^{(i)} \epsilon_{m,t-i}^2 \end{bmatrix} \\
&+ \sum_{j=1}^{p} \begin{bmatrix} b_{11}^{(j)} h_{11,t-i} & \dots & b_{1m}^{(j)} h_{1m,t-i} \\ \vdots & & \vdots \\ b_{1m}^{(j)} h_{1m,t-i} & \dots & b_{mm}^{(j)} h_{mm,t-i}^2 \end{bmatrix} \\
&:= \Omega + \sum_{i=1}^{q} \mathbf{diag}(\epsilon_{t-i}) A^{(i)} \mathbf{diag}(\epsilon_{t-i}) + \sum_{j=1}^{p} B^{(j)} \odot H_{t-j}
\end{aligned}
$$

- No interaction between the different components ($H_t(i,j)$ only function of $\{\epsilon_{i,u}\epsilon_{j,u}, u < t\}$)
- Not stable by aggregation
- Non explicit positivity conditions

Vector GARCH model (VEC-GARCH)

Engle, Kroner (1995)

$$\begin{aligned} \boldsymbol{\epsilon}_t &= \boldsymbol{H}_t^{1/2}\boldsymbol{\eta}_t, \quad \text{where } \boldsymbol{H}_t \text{ is positive definite,} \\ \text{vech}(\boldsymbol{H}_t) &= \boldsymbol{\omega} + \sum_{i=1}^{q} \boldsymbol{A}^{(i)}\text{vech}(\boldsymbol{\epsilon}_{t-i}\boldsymbol{\epsilon}'_{t-i}) + \sum_{j=1}^{p} \boldsymbol{B}^{(j)}\text{vech}(\boldsymbol{H}_{t-j}) \end{aligned}$$

- The most direct generalization of univariate GARCH
- Everything is explained by everything
- The diagonal model is a particular VEC-GARCH (diagonal matrices $A^{(i)}$ and $B^{(j)}$)
- Stability by aggregation
- Positivity conditions are difficult to obtain

BEKK-GARCH model

Engle and Kroner (1995) (Baba, Engle, Kraft and Kroner)

$$\begin{cases} \boldsymbol{\epsilon}_t = \boldsymbol{H}_t^{1/2}\boldsymbol{\eta}_t, & (\boldsymbol{\eta}_t) \text{ iid } (\mathbf{0}, \boldsymbol{I}) \\ \boldsymbol{H}_t = \boldsymbol{\Omega} + \sum_{i=1}^{q}\sum_{k=1}^{K} \boldsymbol{A}_{ik}\boldsymbol{\epsilon}_{t-i}\boldsymbol{\epsilon}'_{t-i}\boldsymbol{A}'_{ik} + \sum_{j=1}^{p}\sum_{k=1}^{K} \boldsymbol{B}_{jk}\boldsymbol{H}_{t-j}\boldsymbol{B}'_{jk} \end{cases}$$

- Positivity conditions are simple ($\boldsymbol{\Omega}$ positive definite). Identifiability of a BEKK representation requires additional constraints.
- Stability by aggregation
- Coefficients of a BEKK representation are difficult to interpret

Constant Conditional Correlation (CCC) model

Bollerslev (1990) ; Extended CCC by Jeantheau (1998)

$$\underline{\boldsymbol{h}}_t = \begin{pmatrix} h_{11,t} \\ \vdots \\ h_{mm,t} \end{pmatrix}, \quad \boldsymbol{D}_t = \text{diag}\left(h_{11,t}^{1/2}, \dots, h_{mm,t}^{1/2}\right), \quad \underline{\boldsymbol{\epsilon}}_t = \begin{pmatrix} \epsilon_{1t}^2 \\ \vdots \\ \epsilon_{mt}^2 \end{pmatrix}.$$

$$\begin{cases} \boldsymbol{\epsilon}_t & = & \boldsymbol{H}_t^{1/2}\boldsymbol{\eta}_t, \qquad \boldsymbol{H}_t = \boldsymbol{D}_t\boldsymbol{R}\boldsymbol{D}_t, \quad \boldsymbol{R} : \text{correlation matrix} \\ \underline{\boldsymbol{h}}_t & = & \underline{\boldsymbol{\omega}} + \sum_{i=1}^{q} \mathbf{A}_i \underline{\boldsymbol{\epsilon}}_{t-i} + \sum_{j=1}^{p} \mathbf{B}_j \underline{\boldsymbol{h}}_{t-j} \end{cases}$$

- Simple condition ensuring the positive definiteness of H_t : the coefficients of the matrices $\mathbf{A}_i$ and $\mathbf{B}_j$ are nonnegative
- No stability by aggregation, no Vec representation.
- The assumption of constant conditional correlations can be too restrictive

Dynamic Conditional Correlation (DCC) model

Engle (2002)

$$H_t = D_t R_t D_t, \qquad R_t = (\operatorname{diag} Q_t)^{-1/2} Q_t (\operatorname{diag} Q_t)^{-1/2},$$

where $\eta_t^* = D_t^{-1}\epsilon_t$ and

$$Q_t = (1-\alpha-\beta)S + \alpha \eta_{t-1}^* \eta_{t-1}^{*'} + \beta Q_{t-1},$$

where $\alpha, \beta \geq 0, \alpha+\beta < 1$, S is a correlation matrix

- Incorrect interpretation of S as $\mathrm{Var}(\eta_t^*)$ and Q_t as $\mathrm{Var}_{t-1}(\eta_t^*)$.

Dynamic Conditional Correlation (DCC) model

Corrected DCC (Aielli (2013)

$$\boldsymbol{Q}_t = (1-\alpha-\beta)\boldsymbol{S} + \alpha \boldsymbol{Q}_{t-1}^{*1/2}\boldsymbol{\eta}_{t-1}^{*}\boldsymbol{\eta}_{t-1}^{*'}\boldsymbol{Q}_{t-1}^{*1/2} + \beta\boldsymbol{Q}_{t-1},$$

where $\boldsymbol{Q}_t^* = \mathrm{diag}\,(\boldsymbol{Q}_t)$.

- $\boldsymbol{S} = E\boldsymbol{Q}_t = E\boldsymbol{Q}_{t-1}^{*1/2}\boldsymbol{\eta}_{t-1}^{*}\boldsymbol{\eta}_{t-1}^{*'}\boldsymbol{Q}_{t-1}^{*1/2}$
- Identifiability constraint : diag$(\boldsymbol{S}) = \boldsymbol{I}_m$.
- Parcimony but the $m(m-1)/2$ conditional correlations have the same dynamic structure.
- cDCC-BEKK :

$$\boldsymbol{Q}_t = \boldsymbol{C} + \sum_{i=1}^{q}\sum_{k=1}^{K}\boldsymbol{A}_{ik}\boldsymbol{Q}_{t-i}^{*1/2}\boldsymbol{\eta}_{t-i}^{*}\boldsymbol{\eta}_{t-i}^{*'}\boldsymbol{Q}_{t-i}^{*1/2}\boldsymbol{A}_{ik}' + \sum_{j=1}^{p}\sum_{k=1}^{K}\boldsymbol{B}_{jk}\boldsymbol{Q}_{t-j}\boldsymbol{B}_{jk}'.$$

Regime Switching DCC model

Pelletier (2006)

$$\boldsymbol{\epsilon}_t = \boldsymbol{H}_t^{1/2}\boldsymbol{\eta}_t, \quad \boldsymbol{H}_t = \boldsymbol{D}_t\boldsymbol{R}_t\boldsymbol{D}_t, \qquad \boldsymbol{R}_t = \boldsymbol{R}(\Delta_t)$$

where (Δ_t) is a Markov chain on $\{1, \dots, d\}$, which is independent of (η_t).

- The model imposes constancy of the correlations within a regime.
- Stochastic CC model : $\boldsymbol{H}_t$ is not the conditional variance of $\boldsymbol{\epsilon}_t$.

Dynamic factor models

- In the Arbitrage Pricing Theory of Ross (1996), returns are generated by a number of common unobserved components, or factors.
- Main argument for using factor models : parsimony.

Factor model of Engle, Ng and Rothschild (1990)

$$\boldsymbol{\epsilon}_t = \boldsymbol{B}\boldsymbol{f}_t + \boldsymbol{\xi}_t, \quad \boldsymbol{\xi}_t \overset{iid}{\sim} (\boldsymbol{0}, \boldsymbol{\Omega}), \quad \boldsymbol{f}_t \in \mathbb{R}^K, \quad K < m$$

where $\boldsymbol{B}$ is a $m \times K$ matrix of factor loadings, $\boldsymbol{f}_t$ is a vector of K factors. Thus, assuming that the factors $f_{k,t}$ are conditionally uncorrelated,

$$\boldsymbol{H}_t = \boldsymbol{\Omega} + \sum_{k=1}^{K} \boldsymbol{\beta}_k \boldsymbol{\beta}_k' \lambda_{k,t},$$

where $\boldsymbol{\Omega}$ is a positive semi-definite matrix, $\boldsymbol{\beta}_k$ are $m \times 1$ vectors of weights, the $\lambda_{k,t}$'s are the factors volatilities with a GARCH structure :

$$\lambda_{kt} = \omega_k + \alpha_k f_{k,t-1}^2 + \gamma_k \lambda_{k,t-1}.$$

Conditionally Orthogonal GARCH models

Based on the singular value decomposition of a symmetric positive definite $m \times m$ matrix V :

$$\boldsymbol{V} = \boldsymbol{P}\boldsymbol{\Lambda}\boldsymbol{P}', \quad \boldsymbol{\Lambda} = \mathrm{diag}\left(\lambda_1, \ldots, \lambda_m\right),$$

where $\boldsymbol{P}$ is orthonormal ($\boldsymbol{P}\boldsymbol{P}' = \boldsymbol{I}_m$).

There exist m principal components which are uncorrelated, and whose variances are the λ_i's.

Conditionally Orthogonal GARCH models

O-GARCH (Alexander and Chibumba (1997), Alexander (2001)) :

$$\boldsymbol{H}_t = \boldsymbol{P}\boldsymbol{\Lambda}_t\boldsymbol{P}', \quad \boldsymbol{\Lambda}_t = \text{diag}\left(\lambda_{1,t}, \ldots, \lambda_{m,t}\right),$$

and $\boldsymbol{P}$ is an orthonormal matrix of eigenvectors of $\boldsymbol{H}_t$, and $\boldsymbol{\Lambda}_t$ is the diagonal matrix of eigenvalues. The λ_{it} are the conditional variances of the eigenvectors, and can be modelled by univariate GARCH.

- Factor interpretation : $\boldsymbol{\epsilon}_t = \boldsymbol{P}\boldsymbol{f}_t$, where $\boldsymbol{f}_t$ is a vector of conditionally uncorrelated factors (with $K = m$: full-factor GARCH).
- In the GO-GARCH extension (Van der Weide (2002)), the matrix $\boldsymbol{P}$ is any nonsingular matrix.

Conditionally Orthogonal GARCH models

$$\boldsymbol{H}_t = \boldsymbol{P}\boldsymbol{\Lambda}_t\boldsymbol{P}', \quad \boldsymbol{\Lambda}_t = \mathrm{diag}\left(\lambda_{1,t}, \ldots, \lambda_{m,t}\right), \quad \boldsymbol{P}\boldsymbol{P}' = \boldsymbol{I}_m.$$

A crucial assumption is that the principal axes are constant over time. This can be

- an advantage because $E(\boldsymbol{H}_t) = \boldsymbol{P}E(\boldsymbol{\Lambda}_t)\boldsymbol{P}'$.
  The loading matrix $\boldsymbol{P}$ can be estimated from the sample covariance matrix of the vector of returns. Univariate GARCH models can be estimated in a 2nd step on the principal components.
- Also a strong restriction. Extensions require a dynamic model for $\boldsymbol{P}_t$ (Aielli and Caporin (2015)).

Cholesky GARCH model

Darolles, Francq and Laurent (2018)

Cholesky theorem : Any symmetric positive-definite matrix Σ can be written $\Sigma = LL'$ where L is lower triangular.

Alternative decomposition : $A = LDL'$ where D is diagonal and L is lower triangular with ones on the diagonal.

Cholesky GARCH :

$$\boldsymbol{H}_t = \boldsymbol{L}_t \boldsymbol{D}_t \boldsymbol{L}_t', \quad \boldsymbol{D}_t = \text{diag}\left(h_{11,t}, \dots, h_{mm,t}\right),$$

where L_t is lower triangular with ones on the diagonal.

- No positivity constraints on the coefficients of L_t.
- Factor model interpretation.

Cholesky Decomposition of $\mathbf{\Sigma} = \mathrm{Var}\left(\boldsymbol{\epsilon}\right)$

$\boldsymbol{\epsilon} = (\epsilon_1, \ldots, \epsilon_m)'$

Letting $v_1 := \epsilon_1$, we have

$$\epsilon_2 = \ell_{21} v_1 + v_2 = \beta_{21}\epsilon_1 + v_2,$$

where $\beta_{21} = \ell_{21}$ is the beta in the regression of ϵ_2 on ϵ_1, and v_2 is orthogonal to ϵ_1. Recursively, we have

$$\epsilon_i = \sum_{j=1}^{i-1} \ell_{ij} v_j + v_i = \sum_{j=1}^{i-1} \beta_{ij}\epsilon_j + v_i, \qquad \text{for } i = 2, \ldots, m,$$

where v_i is uncorrelated to $v_1, \ldots, v_{i-1}$, and thus uncorrelated to $\epsilon_1, \ldots, \epsilon_{i-1}$. In general the order of the series matters

Cholesky Decomposition of $\boldsymbol{\Sigma} = \text{Var}(\boldsymbol{\epsilon})$

In matrix form,

$$\boldsymbol{\epsilon} = \boldsymbol{L}\boldsymbol{v} \qquad \text{and} \qquad \boldsymbol{v} = \boldsymbol{B}\boldsymbol{\epsilon},$$

where $\boldsymbol{L}$ and $\boldsymbol{B} = \boldsymbol{L}^{-1}$ are lower unitriangular and $\boldsymbol{G} := \text{var}(\boldsymbol{v})$ is diagonal. We obtain the Cholesky decomposition $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{G}\boldsymbol{L}'$ or $\boldsymbol{\Sigma}^{-1} = \boldsymbol{B}'\boldsymbol{G}^{-1}\boldsymbol{B}$. Conditioning on $\mathcal{F}_{t-1}$, $\boldsymbol{\Sigma}_t = \boldsymbol{L}_t\boldsymbol{G}_t\boldsymbol{L}_t'$ and $\boldsymbol{\Sigma}_t^{-1} = \boldsymbol{B}_t'\boldsymbol{G}_t^{-1}\boldsymbol{B}_t$. We thus need
- a diagonal ARCH-type model for the factors vector $\boldsymbol{v}_t$
- a time series model for $\boldsymbol{L}_t$ (or $\boldsymbol{B}_t$), without particular constraint.

Example : $\boldsymbol{\Sigma} = \boldsymbol{LGL'} = \boldsymbol{DRD}$, $m = 2$

Letting

$$g_1 = \sigma_1^2 = \text{var}(\epsilon_1), \quad \epsilon_2 = \ell_{21}\epsilon_1 + v_2, \quad g_2 = \text{var} v_2, \quad \sigma_2^2 = \text{var}(\epsilon_2)$$

and $\rho = \text{cor}(\epsilon_1, \epsilon_2)$, we have

$$\boldsymbol{L} = \begin{bmatrix} 1 & 0 \\ \ell_{21} & 1 \end{bmatrix}, \quad \boldsymbol{G} = \begin{bmatrix} g_1 & 0 \\ 0 & g_2 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}, \quad \boldsymbol{R} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} g_1 & \ell_{21}g_1 \\ \ell_{21}g_1 & \ell_{21}^2 g_1 + g_2 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Positivity constraints : $g_1 > 0$ and $g_2 > 0$ or $\sigma_1^2 > 0$, $\sigma_1^2 > 0$ and $\rho^2 < 1$.

Example : $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{G}\boldsymbol{L}'$, $\boldsymbol{\Sigma}^{-1} = \boldsymbol{B}'\boldsymbol{G}^{-1}\boldsymbol{B}$, $m = 3$

$$\boldsymbol{L} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 0 & 0 \\ -\beta_{21} & 1 & 0 \\ -\beta_{31} & -\beta_{32} & 1 \end{bmatrix}, \quad \boldsymbol{G} = \begin{bmatrix} g_1 & 0 & 0 \\ 0 & g_2 & 0 \\ 0 & 0 & g_3 \end{bmatrix},$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} g_1 & \ell_{21}g_1 & \ell_{31}g_1 \\ \ell_{21}g_1 & \ell_{21}^2 g_1 + g_2 & \ell_{21}l_{31}g_1 + \ell_{32}g_2 \\ \ell_{31}g_1 & \ell_{21}\ell_{31}g_1 + \ell_{32}g_2 & \ell_{31}^2 g_1 + \ell_{32}^2 g_2 + g_3 \end{bmatrix}.$$

Remark : In $\boldsymbol{\Sigma} = \boldsymbol{D}\boldsymbol{R}\boldsymbol{D}$ the constraints on the elements of $\boldsymbol{R}$ are

$$\rho_{12}^2 + \rho_{13}^2 + \rho_{23}^2 - 2\rho_{12}\rho_{13}\rho_{23} \leq 1.$$

In $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{G}\boldsymbol{L}'$ there is no constraint on the ℓ_{ij}'s, but just $g_i > 0$ for $i = 1, \dots, m$.

Existing results

Stationarity results for MGARCH models exist for certain classes only :

- The CCC
- The BEKK
- Some DCC models
- Some factor models

Incorrect results also exist.

The VARMA representation is useless for stationarity

Letting $\nu_t = \text{vech}(\epsilon_t \epsilon_t') - \text{vech}(H_t)$ we obtain the VARMA

$$\text{vech}(\epsilon_t \epsilon_t') = \omega + \sum_{i=1}^{r} C^{(i)} \text{vech}(\epsilon_{t-i} \epsilon_{t-i}') + \nu_t - \sum_{j=1}^{p} B^{(j)} \nu_{t-j}.$$

In the literature, one finds the argument that the model is weakly stationary if the roots of $\det \left(I_s - \sum_{i=1}^{r} C^{(i)} z^i \right)$ are outside the unit circle ($s = m(m+1)/2$).

Not correct since

$$\text{vech}(\epsilon_t \epsilon_t') = \left(\sum_{i=1}^{r} C^{(i)} B^i \right)^{-1} \left(\omega + \nu_t - \sum_{j=1}^{p} B^{(j)} \nu_{t-j} \right)$$

constitutes a solution only if ν_t is a function of $\{\eta_{t-u}, u > 0\}$.

Stationarity of CCC GARCH models

Letting $\tilde{\eta}_t = R^{1/2}\eta_t$, we get

$$\underline{\epsilon}_t = \Upsilon_t \underline{h}_t, \quad \text{where} \quad \Upsilon_t = \begin{pmatrix} \tilde{\eta}_{1t}^2 & 0 & \dots & 0 \\ 0 & \ddots & & \\ \vdots & & \ddots & \\ 0 & \dots & & \tilde{\eta}_{mt}^2 \end{pmatrix}.$$

Multiplying by Υ_t the equation of $\underline{h}_t$ we thus have

$$\underline{\epsilon}_t = \Upsilon_t \underline{\omega} + \sum_{i=1}^{q} \Upsilon_t \mathbf{A}_i \underline{\epsilon}_{t-i} + \sum_{j=1}^{p} \Upsilon_t \mathbf{B}_j \underline{h}_{t-j}$$

which can be written under the form

$$\underline{z}_t = \underline{b}_t + \boldsymbol{C}_t \underline{z}_{t-1},$$

where $\underline{z}_t = \left(\underline{\epsilon}_t', \dots, \underline{\epsilon}_{t-q+1}', \underline{h}_t', \dots, \underline{h}_{t-p+1}'\right)' \in \mathbb{R}^{m(p+q)}$.

Stationarity of CCC GARCH models

Strict stationarity of the CCC model

A necessary and sufficient condition for the existence of a strictly stationary and non anticipative solution process to the CCC Model is

$$\gamma < 0,$$

where γ is the top Lyapunov exponent of the sequence $(\boldsymbol{C}_t)$. This stationary and non anticipative solution, when $\gamma < 0$, is unique and ergodic.

Stationarity of the BEKK model

Boussama, Fuchs, Stelzer (2011)

Let

$$\boldsymbol{z}_t = (\boldsymbol{\epsilon}_t', \ldots, \boldsymbol{\epsilon}_{t-q+1}', \mathsf{vech}(\boldsymbol{H}_t)', \ldots, \mathsf{vech}(\boldsymbol{H}_{t-p+1})')'$$

It can be shown (it's not easy) that : if
1. the law of $\boldsymbol{\eta}_t$ admits a positive density on $\mathbb{R}^m$,
2. $\rho\left(\sum_{i=1}^{q} \boldsymbol{A}^{(i)} + \sum_{j=1}^{p} \boldsymbol{B}^{(j)}\right) < 1$,

then the BEKK model admits a unique strictly stationary and non anticipative solution, which is also second-order stationary.

Stationarity of the cDCC-BEKK model

Aielli (2013)

$$\boldsymbol{H}_t = \boldsymbol{D}_t \boldsymbol{R}_t \boldsymbol{D}_t, \qquad \boldsymbol{R}_t = (\boldsymbol{Q}_t^*)^{-1/2} \boldsymbol{Q}_t (\boldsymbol{Q}_t^*)^{-1/2},$$

where

$$\boldsymbol{Q}_t = \boldsymbol{C} + \sum_{i=1}^{q} \sum_{k=1}^{K} \boldsymbol{A}_{ik} \boldsymbol{Q}_{t-i}^{*1/2} \boldsymbol{\eta}_{t-i}^{*} \boldsymbol{\eta}_{t-i}^{*'} \boldsymbol{Q}_{t-i}^{*1/2} \boldsymbol{A}_{ik}' + \sum_{j=1}^{p} \sum_{k=1}^{K} \boldsymbol{B}_{jk} \boldsymbol{Q}_{t-j} \boldsymbol{B}_{jk}'.$$

A consequence of the stationarity of the BEKK model.

Under the stationarity conditions of the BEKK model, and **if** $\boldsymbol{D}_t = \boldsymbol{D}(\boldsymbol{\eta}_{t-1}^*, \ldots)$, the cDCC-BEKK model admits a unique strictly stationary and non anticipative solution, which is also second-order stationary.

Stationarity of Cholesky GARCH models

$$\boldsymbol{H}_t = \boldsymbol{L}_t \boldsymbol{D}_t \boldsymbol{L}_t', \quad \boldsymbol{D}_t = \mathrm{diag}\left(h_{11,t}, \dots, h_{mm,t}\right),$$

where L_t is lower triangular with ones on the diagonal.

$$\boldsymbol{\epsilon}_t = \boldsymbol{L}_t \boldsymbol{u}_t, \qquad \boldsymbol{u}_t = \boldsymbol{D}_t^{1/2} \boldsymbol{\eta}_t$$

A strictly stationary solution is obtained by specifying

- A strictly stationary model for $\boldsymbol{u}_t$ (e.g.a CCC)
- Specifying $\boldsymbol{L}_t$ as a function of the past of $\boldsymbol{u}_t$.